Person-Level Kinship Verification: Leakage-Free Evaluation, Set-Based Aggregation, and a Negative Result for Quantum-Inspired Metrics

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***Abstract*—Facial kinship verification is evaluated almost exclusively as a photograph-versus-photograph decision on four benchmarks. We argue that this formulation is both harder to measure reliably and weaker than the data supports. First, we quantify the leakage admitted by conventional splitting: because Families In the Wild draws $57{,}120$ pairs from 95 families and 472 identities, a random pair-level split places an identity from *every* test pair into the training set, a violation we measure at $11{,}424/11{,}424$ (100%). We propose a leakage-free protocol based on grouped k-fold cross-validation in which every family is tested exactly once and negatives are regenerated within each fold; this reduces seed-to-seed accuracy variation from 22.5 to 12.1 points. Second, we show that the conventional formulation discards most of the available evidence: FIW supplies a median of five photographs per identity, yet the standard protocol scores one against one. Representing each *person* by their photograph set, as template-based face recognition does, improves ROC-AUC by $+0.077$ on FIW under the leakage-free protocol, while reducing exactly to the single-image case on corpora storing one photograph per person. We further show that scoring father–mother–child triads jointly, rather than as independent pairs, adds $+0.032$ ROC-AUC on TSKinFace ($p = 0.0001$). Finally we report a systematic negative result: nine quantum-inspired formulations, spanning similarity measures, feature extractors and inductive biases, each evaluated against a capacity-matched classical control, fail to improve on their controls, and four are significantly worse. Each of the nine reads its decision out of the quantum state as a single scalar; a companion study shows that this readout is itself a substantial bottleneck, so the negative result should be read as scoped to scalar-readout formulations rather than to quantum-inspired representations in general. We further show that both findings replicate under a second backbone (ArcFace), where the person-level gain is larger still ($+0.098$), and we document a subgroup artefact of independent methodological interest: an apparent 0.28 ROC-AUC deficit for mother–son pairs proves to be a single-family effect that disappears (0.031 spread) once no family is allowed to dominate a fold. All protocol and evaluation code, together with the JSON artefacts underlying every figure and table, is released.**

Code and data-processing scripts: https://github.com/svaibhav-7/Quantum_kinship_verification

***Index Terms*—Facial kinship verification, evaluation protocol, dataset bias, data leakage, quantum-inspired machine learning, negative results.**

I. Introduction

Facial kinship verification asks whether two facial images depict biologically related individuals. Progress is measured on a small set of public benchmarks, and reported accuracies have risen steadily. This paper examines whether the evaluation protocols supporting those numbers are sound.

We make three contributions, of which the first two concern how the task is posed and measured rather than which model is applied to it.

A leakage-free protocol, and the leakage it removes. Splitting a corpus at the level of *pairs* rather than *identities* does not isolate a test set. FIW draws $57{,}120$ pairs from only 95 families and 472 identities, so identities recur across pairs an average of 242 times; under a random pair-level split we measure that 100% of test pairs share an identity with training. Correcting this is less direct than it appears, because every negative pair joins two families and, treating families as vertices, yields a graph with a single connected component over all of FIW: no family-disjoint partition of the existing pairs exists, and negatives must be regenerated within each fold. That kinship datasets carry construction artefacts is known — prior work has shown that negatives assembled by recombining family members admit shortcuts [1] — but we are not aware of a quantification of identity leakage under the splitting practice in common use, nor of a released protocol that removes it.

Person-level rather than photograph-level verification. The conventional formulation compares two photographs. FIW supplies a median of five photographs per identity, so this discards most of the evidence about each person. Template-based protocols in face recognition score a person represented by an image set [2]. Image sets have been used in kinship verification to learn a set-to-set metric [12]; what we evaluate is different and simpler — aggregating a subject's photographs into a single template scored by an ordinary pairwise decision — and, to our knowledge, it has not been measured under a leakage-free protocol. Doing so improves ROC-AUC by $+0.077$ on FIW, and by construction leaves corpora with one photograph per person unchanged. We additionally show that father–mother–child triads carry information that pairwise decomposition destroys, worth a further $+0.032$ ROC-AUC.

A negative result for quantum-inspired metrics. Quantum-inspired similarity has been proposed for this task. We evaluate nine formulations against capacity-matched classical controls; none improves on its control and four are significantly worse. We give a structural account of why, rather than attributing the outcome to implementation.

Concretely we contribute: (i) a measured quantification of identity leakage under conventional splitting, and a released protocol that removes it; (ii) person-level and triadic formulations of the task, with gains measured under that protocol; and (iii) a pre-registered nine-formulation negative result for scalar-readout quantum-inspired metrics.

II. Related Work

A. Kinship verification

Early work framed the task as metric learning, learning a projection under which related faces lie closer than unrelated ones [3]. Subsequent methods replaced the hand-designed metric with deep pairwise encoders and, more recently, attention- and graph-based architectures [4]. Evaluation is concentrated on four corpora: KinFaceW-I and KinFaceW-II [5], TSKinFace [6], and Families In the Wild [7], the last also underpinning the RFIW challenge series [8].

Reported accuracies are obtained under each dataset's official protocol. We therefore do not tabulate them against our results: Section IV shows that those protocols admit two forms of optimistic bias, so a direct ranking would mislead in both directions.

B. Evaluation bias in vision benchmarks

That benchmark protocols can inflate reported performance is established in adjacent fields. Face recognition distinguishes subject-disjoint from random splitting precisely because identity overlap leaks information [9], and template-based protocols such as IJB-B/C score a *person* represented by several images rather than a single photograph [2]. More generally, models exploit incidental correlations in preference to the intended signal, a failure mode catalogued as shortcut learning [10] and, earlier, as dataset bias [11].

Both concerns apply directly to kinship verification. Construction artefacts in kinship corpora have been reported [1], but we are not aware of a quantification of identity leakage under the splitting practice in common use, nor of a person-level protocol for this task: existing multi-image work addresses image-set subspace representations or matching within family photographs, which is a different problem from aggregating a subject's photographs into a single template for pairwise kinship decisions [12]. Sections IV and VIII address both gaps.

C. Quantum-inspired machine learning

Quantum feature maps embed classical data in a Hilbert space where inner products act as kernels [13], with reported gains on classical tasks [14]. Related lines use SWAP-test state fidelity as a similarity measure and density matrices as classifiers. Such methods have been proposed for kinship verification, and the claim that they outperform classical counterparts on classical data is contested; dequantization results show that several purported advantages vanish under scrutiny [15].

Section XI contributes to that debate empirically, evaluating nine formulations against capacity-matched classical controls.

III. Datasets and Experimental Setup

We use the four corpora on which facial kinship verification is conventionally evaluated. Table I characterises them by the two properties that govern this work: how many grouping units are available for a leakage-free split, and how many photographs each stores per individual.

Two features drive the rest of the paper. FIW draws $57{,}120$ pairs from only 95 families and 472

TABLE I
THE FOUR BENCHMARK CORPORA. PHOTOGRAPHS PER IDENTITY IS THE PROPERTY THAT DETERMINES WHETHER PERSON-LEVEL AGGREGATION IS POSSIBLE AT ALL.

Corpus	Pairs	Families	Identities	Photos/id.
KinFaceW-I	1,066	533	1,066	1.00
KinFaceW-II	2,000	1,000	2,000	1.00
TSKinFace	2,236	559	1,677	1.00
FIW	57,120	95	472	5.28

TABLE II
IDENTITY LEAKAGE UNDER A RANDOM 80/20 PAIR-LEVEL SPLIT OF FIW.

Quantity	Value
Pairs	57,120
Distinct families	95
Distinct identities	472
Mean appearances per identity	242
Largest family, share of all pairs	19%
Test pairs sharing an identity with train	11,424 / 11,424 (100%)
Test pairs whose family was seen in train	11,424 / 11,424 (100%)

identities — 121 pairs per identity — which is what makes pair-level splitting leak so completely. And only FIW supplies multiple photographs per individual, so it is the sole corpus on which person-level aggregation can differ from the conventional formulation. We report per corpus rather than averaging, because an average over these four would conceal exactly that asymmetry.

All experiments use a single frozen backbone: FaceNet (Inception-ResNet-v1) pretrained on VGGFace2, producing 512-dimensional ℓ_2-normalised embeddings. Holding it fixed isolates the protocol and representation questions from representation-learning effects, and bounds the claims accordingly (Section XII).

Every numeric claim derives from JSON artefacts written by the evaluation scripts, and each experiment is deterministic given its seed: repeated runs reproduce the reported figures to four decimal places. The implementation carries 138 tests, including assertions that no grouping unit spans a fold boundary.

IV. SOURCES OF OPTIMISTIC BIAS

A. Identity leakage under pair-level splitting

Let $\mathcal{P}$ be a set of labelled pairs and $\mathrm{id}(\cdot)$ the identity depicted in an image. A split $(\mathcal{P}_{\mathrm{tr}}, \mathcal{P}_{\mathrm{te}})$ is *identity-disjoint* if

$$\Big(\bigcup_{p\in\mathcal{P}_{\mathrm{tr}}} \mathrm{id}(p)\Big) \cap \Big(\bigcup_{p\in\mathcal{P}_{\mathrm{te}}} \mathrm{id}(p)\Big) = \emptyset. \tag{1}$$

Random splitting over $\mathcal{P}$ does not enforce this. Table II reports the measured violation on FIW.

B. A non-obvious obstacle to correcting it

Splitting on families is not sufficient by itself. Each *negative* pair joins two families, and treating families as vertices and negatives as edges yields a graph over FIW with a single connected component containing all 95 families. No family-disjoint partition of the existing pairs exists. Negatives must therefore be discarded and regenerated *within* each side after the split. We found this empirically: a splitter that passed synthetic unit tests failed on the real corpus for exactly this reason.

A second consequence is data loss. Because preexisting negatives straddle any group boundary, naively dropping them removes usable data — on KinFaceW-I, 173 of $1{,}066$ pairs, leaving only 119 test pairs. Regenerating negatives per side instead recovers the test set to 212 pairs on KinFaceW-I and 400 on KinFaceW-II.

C. Family membership as a predictor in TSKinFace

That kinship benchmarks carry construction artefacts is established. Prior work has shown that negatives assembled by recombining members of family photographs permit a model to succeed without learning kinship [1], and the TSKinFace protocol documents its own negative-construction procedure [6]. We do not claim to be the first to observe that this dataset family is vulnerable.

The specific regularity we measure is distinct from the same-photograph effect, and worth separating. TSKinFace stores one cropped face per family member, so *no* pair — positive or negative — shares a source photograph, and the same-photograph shortcut cannot arise. What does hold is that all 800 positives are drawn within a family and all 800 negatives across families, so family membership predicts the label perfectly. A model can exploit correlated capture conditions — illumination, camera, background — shared by members photographed together.

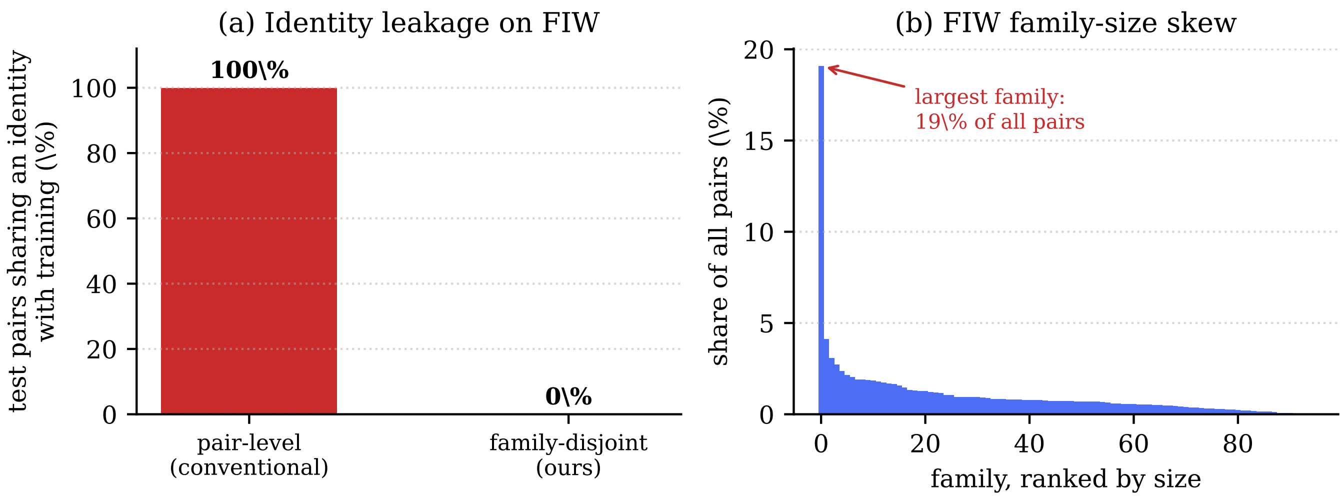

Fig. 1. (a) Identity leakage on FIW: under conventional pair-level splitting every test pair shares an identity with training, which the family-disjoint protocol reduces to zero. (b) The family-size distribution that makes a single split unstable — the largest family alone accounts for 19% of all pairs.

TABLE III
TSKinFace pair construction. Under the standard protocol family membership predicts the label perfectly; no pair shares a source photograph in either construction.

Construction	Same-fam. pos.	Same-fam. neg.	Cosine AUC
Standard	100%	0%	0.7278
Shortcut-free (ours)	100%	50%	0.6880

To measure the margin this affords, we rebuild the negatives so that half are drawn *within* a family, pairing father against mother: co-photographed, hence sharing every session cue, but not kin.

The 0.0398 ROC-AUC reduction is measured on raw embeddings without training, so it reflects separability available to any model rather than a property of ours. All TSKinFace results in this paper use the shortcut-free construction.

We check separately that our FIW results do not rest on the same-photograph effect that prior work identifies: 8.2% of FIW positives are drawn from a single source photograph (against 0% of negatives), and excluding them changes accuracy by $+0.08$ points ($82.53\% \rightarrow 82.61\%$). The effect is present in FIW but our model does not exploit it.

V. Corrected Protocol

We adopt grouped k-fold cross-validation with $k = 5$. The grouping unit is the family, which is the unit that shares identities and capture conditions.

1) **Positives only are partitioned.** Families are dealt largest-first into the currently smallest fold, which keeps folds balanced despite the highly skewed family sizes.
2) **Negatives are generated per side**, after partitioning, from the families present on that side.
3) **Group keys are dataset-scoped.** The family identifier `fd_003` occurs in both KinFaceW-I and KinFaceW-II denoting different families; an unscoped key merges them and corrupts the disjointness check.
4) **Family contribution is capped** at $1{,}500$ pairs, preventing the largest FIW family (19% of all pairs) from dominating whichever fold it enters.
5) **Thresholds are calibrated on a group-disjoint validation slice** of the training side. The test fold is scored once.

Fold integrity is verified directly rather than assumed: zero folds exhibit train/test group overlap, and every family is tested exactly once with no duplicates ($95/95$ FIW, $533/533$ KinFaceW-I, $1000/1000$ KinFaceW-II, $559/559$ TSKinFace).

A. *Effect on measurement stability*

Table IV shows why a single group-disjoint split is insufficient. Accuracy on one split varies by up to 14.2 points with the random seed, because FIW's test side reduces to a small number of families of very unequal size.

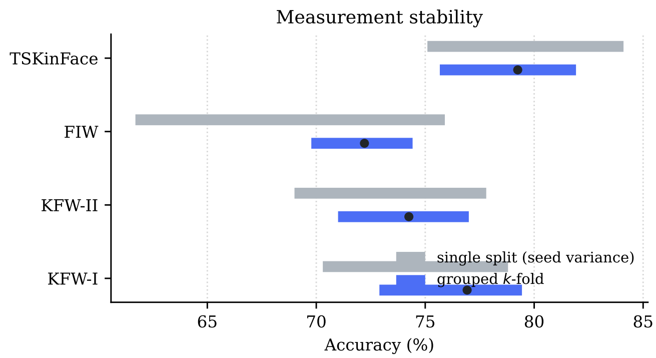

Fig. 2. Accuracy range under a single group-disjoint split across random seeds (grey) against grouped k-fold (blue, mean marked). A single split is too unstable to support a headline number: on FIW the seed-to-seed range spans 14.2 points.

TABLE IV
ACCURACY RANGE ACROSS SEEDS (SINGLE SPLIT) VERSUS ACROSS FOLDS (GROUPED k-FOLD).

Dataset	Single split	Grouped k-fold
FIW	61.7–75.9 (14.2)	69.8–74.4 (4.6)
KinFaceW-I	70.3–78.8 (8.5)	72.9–79.4 (6.5)
KinFaceW-II	69.0–77.8 (8.8)	71.0–77.0 (6.0)
TSKinFace	75.1–84.1 (9.0)	75.7–81.9 (6.2)
Overall spread	22.5	12.1

VI. Reference Model

To measure the protocol rather than an architecture, we use a deliberately conventional model. Frozen FaceNet embeddings (512-d) pass through a shared Siamese encoder; kinship is symmetric, so only symmetric pair features are formed — sum, absolute difference, elementwise product, and cosine — and concatenated with a one-hot relation code before a three-layer MLP. Predictions are averaged over both argument orders, making the model exactly symmetric. The model has $497{,}881$ parameters.

Its plainness is deliberate: findings in Sections IV and V cannot then be attributed to architectural novelty. We note that a linear probe on the same frozen embeddings reaches 0.702 ROC-AUC against this model's 0.810 on FIW, confirming the head performs non-trivial work and that the frozen backbone is not the binding constraint.

VII. Results

Table V reports results under the corrected protocol. We deliberately omit a comparison against published accuracies. Those numbers are obtained under each dataset's official protocol, which permits the leakage and shortcut quantified in Section IV; presenting a ranking against them would mislead in both directions. We regard establishing comparable baselines under the corrected protocol as work the community must do collectively, and we release our implementation to that end.

TABLE V
GROUPED 5-FOLD RESULTS. EVERY FAMILY TESTED EXACTLY ONCE.

Dataset	Accuracy (95% CI)	ROC-AUC
KinFaceW-I	76.92 ± 2.19	0.8752 ± 0.0146
KinFaceW-II	74.25 ± 1.95	0.8320 ± 0.0190
FIW	72.21 ± 1.55	0.8101 ± 0.0175
TSKinFace (shortcut-free)	79.25 ± 2.32	0.8814 ± 0.0192
Mean	**75.66**	**0.8497**

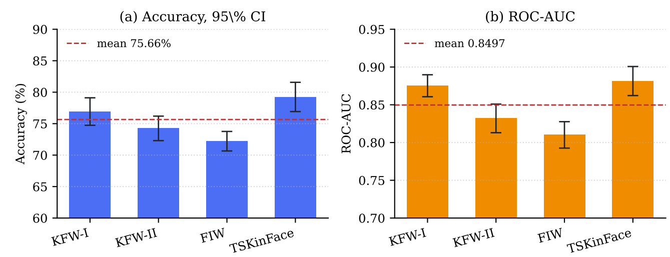

Fig. 3. Grouped 5-fold results, every family tested exactly once. (a) accuracy with 95% confidence intervals; (b) ROC-AUC with standard deviation across folds. Dashed line marks the four-dataset mean.

A. *Per-relation analysis*

VIII. Set-Level and Triadic Representations

Sections IV–VII concern how the task is *measured*. We now examine two ways in which the task is conventionally *posed* that discard available evidence.

A. *Identity photo sets*

The standard protocol scores one photograph against one photograph. FIW, however, supplies a median of five photographs per identity, so the conventional formulation discards most of the evidence for each person. We represent each identity by its photo set and form symmetric set statistics: the ℓ_2-normalised centroid, the within-set spread (1

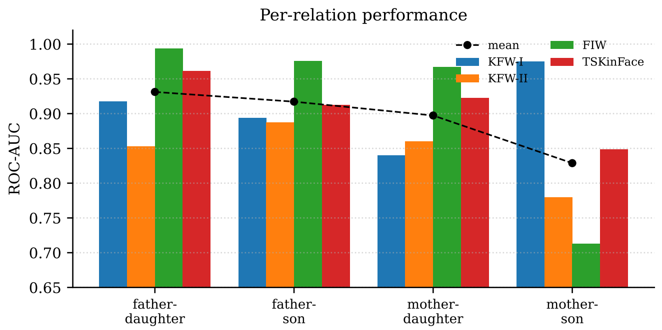

Fig. 4. Per-relation ROC-AUC. Mother–son is the weakest relation on three of four datasets and, on FIW, the largest subset (5,296 pairs).

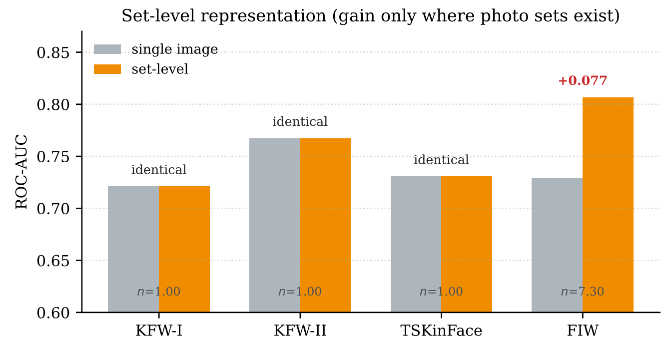

Fig. 5. Set-level representation. The gain appears only where photo sets exist (n = mean images per identity). The three single-photograph corpora are bit-identical to the single-image baseline, which is the degradation guarantee asserted in the test suite.

TABLE VI
SET-LEVEL REPRESENTATION UNDER GROUPED 5-FOLD. GAINS APPEAR ONLY WHERE PHOTO SETS EXIST; SINGLE-PHOTOGRAPH CORPORA ARE BIT-IDENTICAL TO THE BASELINE, AS REQUIRED.

Dataset	Set size	Single	Set-level	Δ
KinFaceW-I	1.00	0.7210	0.7210	0.0000
KinFaceW-II	1.00	0.7672	0.7672	0.0000
TSKinFace	1.00	0.7306	0.7306	0.0000
FIW	7.30	0.7296	**0.8066**	**+0.0771**

TABLE VII
TRIADIC SCORING ON TSKINFACE, GROUPED 5-FOLD, FOLDS ASSIGNED BY THE PARENTS' FAMILY.

Arm	Accuracy (95% CI)	ROC-AUC
Best single parent	67.62 ± 2.69	0.7569
Triadic	$\mathbf{69.05 \pm 2.65}$	**0.7893**
Triadic + phase sweep	69.50 ± 2.41	0.7887

minus mean pairwise cosine), the set size, and the maximum, minimum, mean and standard deviation of the cross-set similarity matrix.

A set of size one must reduce exactly to the single-image case: the centroid is the embedding, the spread is zero, and every cross-set statistic collapses to the single cosine. We assert this in the test suite rather than assume it, which matters because the three remaining benchmarks store exactly one photograph per person.

Table VI reports the outcome. On FIW the representation is worth $+0.0771$ ROC-AUC, closely matching an untrained embedding-level estimate of $+0.071$, which indicates the gain is intrinsic to the representation rather than a product of additional model capacity. The three single-photograph corpora are unchanged to machine precision.

We stress the narrowness of this result: it is a large gain on one benchmark and exactly zero on three, and any aggregate over the four would obscure that. The practical implication is that kinship verification deployments in which several photographs of a person are available — the common case — are leaving a substantial margin unclaimed, and that benchmarks storing a single photograph per identity cannot measure this.

B. Triadic structure

TSKinFace supplies father, mother and child jointly, but the pairwise protocol decomposes each triad into two independent comparisons. We instead score the triad: alongside $\cos(F, C)$ and $\cos(M, C)$ we compute the best convex parental mixture $\max_\alpha \cos\big(\widehat{\alpha F + (1-\alpha)M},\, C\big)$, the maximising α, the gain of that mixture over the better single parent, and the parental similarity $\cos(F, M)$.

The last quantity is not expressible in a pairwise formulation and is the strongest single contributor: a child resembling one parent is weaker evidence of kinship when the two parents already resemble each other.

Triadic scoring improves ROC-AUC by $+0.0324$ over the best single parent (paired t-test over folds, $t = 14.93$, $p = 0.0001$; the gain is positive in all five folds). Negatives retain the true parents and substitute a child from another family, so the label depends on kinship rather than on any property of the parent pair.

IX. EXTENDED ANALYSIS

The corrected protocol makes three further analyses interpretable: where the model's errors concentrate, how the person-level gain scales with the

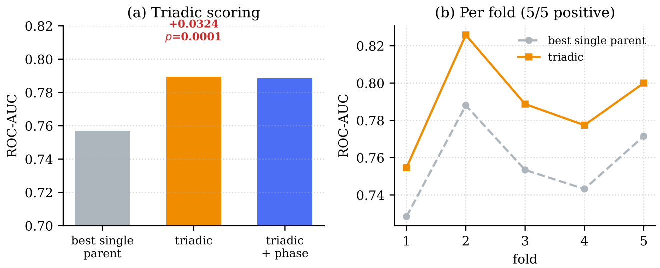

Fig. 6. Triadic scoring on TSKinFace. (a) ROC-AUC by arm; (b) per fold, with the triadic arm ahead in all five.

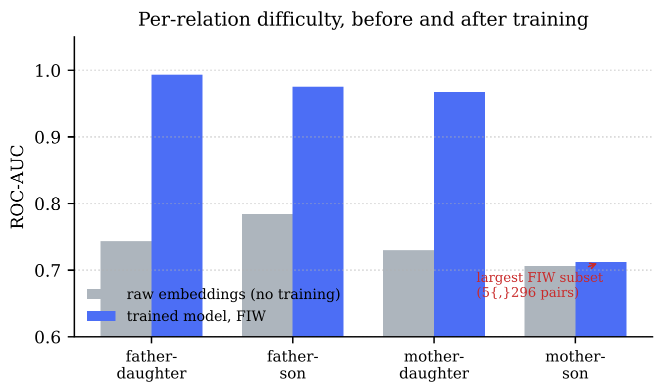

Fig. 7. Per-relation difficulty before and after training. Raw FaceNet embeddings separate the four relations within 0.078 ROC-AUC; the trained model spans 0.281. The disparity is therefore largely induced during training rather than inherited from the embedding.

evidence supplied, and how the score distributions differ across corpora.

A. Where the errors concentrate

Table VIII reports per-relation performance measured on a single held-out fold, the form in which such comparisons are usually presented. Read at face value it suggests a large relation effect: on FIW father–daughter reaches 0.9933 ROC-AUC while mother–son reaches 0.7124. Section IX-B shows this reading is unsound, and we retain the table because the discrepancy between it and the corrected measurement is itself the point.

Figure 7 gives the first indication that the single-fold reading is misleading. On raw embeddings with no training the four relations span only 0.7065–0.7841, a range of 0.078, while the single-fold trained estimate spans 0.281. A representation separating the relations that evenly is unlikely to support a threefold larger disparity after training, and Section IX-B identifies the cause.

B. A correction: mother–son is not the weak relation

An earlier analysis of these results, based on a single held-out fold, reported mother–son as markedly weaker than the other relations (0.712 ROC-AUC against father–daughter's 0.993). Re-examining that fold shows the comparison was not sound: 92% of its mother–son pairs came from a single family, which scores 0.701, while the only other contributing family scores 0.846. The apparent relation effect was a family effect.

Table IX repeats the measurement under grouped k-fold with a per-family cap, so no family can dominate a relation's estimate.

Under the corrected measurement mother–son reaches 0.7910, the second highest of the four, and the relations span 0.0310 ROC-AUC rather than 0.281. That residual spread is close to the 0.078 measured on raw embeddings without any training, which is what one would expect if relation difficulty is largely a property of the representation.

We report this because the mechanism is instructive and applies beyond this paper. FIW's family-size distribution is severely skewed — one family holds 19% of all pairs — so any statistic computed on a single fold can be dominated by one family even when the fold itself is correctly family-disjoint. Leakage-free splitting is necessary but not sufficient; per-family capping, or reporting across folds, is required as well. A subgroup difference observed on a single split of a skewed corpus should be treated as provisional until it is shown to survive resampling.

C. How the person-level gain scales

A second photograph is worth $+0.041$ ROC-AUC on its own — more than half the total available gain — and the curve is substantially flat beyond five, reaching $+0.070$ at ten (Table X, Fig. 8). The practical reading is that a deployment able to obtain roughly five photographs of each person captures nearly all of the benefit.

That the curve is computed on frozen embeddings without training matters for interpretation: the gain is a property of aggregating evidence about a person, not of additional model capacity, consistent with the $+0.077$ obtained under the full protocol.

TABLE VIII

PER-RELATION ROC-AUC. THE SECOND COLUMN IS RAW COSINE SIMILARITY ON FROZEN EMBEDDINGS WITH NO TRAINING; THE REMAINDER ARE THE TRAINED MODEL PER CORPUS. MOTHER–SON IS WEAKEST ON THREE OF FOUR CORPORA WHILE BEING THE LARGEST FIW SUBSET (5,296 PAIRS).

Relation	Raw embedding	KinFaceW-I	KinFaceW-II	FIW	TSKinFace	Mean
Father–daughter	0.7434	0.9174	0.8529	0.9933	0.9612	0.9312
Father–son	0.7841	0.8935	0.8875	0.9754	0.9124	0.9172
Mother–daughter	0.7295	0.8400	0.8598	0.9668	0.9227	0.8973
Mother–son	0.7065	0.9750	0.7795	0.7124	0.8483	0.8288

TABLE IX

PER-RELATION ROC-AUC UNDER GROUPED k-FOLD WITH A PER-FAMILY CAP. THE SINGLE-FOLD COLUMN IS THE EARLIER, UNSOUND ESTIMATE; THE REMAINING COLUMNS ARE THE CORRECTED MEASUREMENT. UNDER A PROTOCOL IN WHICH NO FAMILY DOMINATES, THE FOUR RELATIONS SPAN 0.0310 ROC-AUC.

Relation	Single fold	Mean	SD	Range	Families/fold
Father–daughter	0.9933	0.7635	0.0219	0.7435–0.7953	12.6
Father–son	0.9754	0.7946	0.0752	0.7050–0.8942	12.2
Mother–daughter	0.9668	0.7766	0.0601	0.7147–0.8730	14.4
Mother–son	0.7124	0.7910	0.0381	0.7497–0.8390	12.4

TABLE X

SET-LEVEL ROC-AUC AGAINST PHOTOGRAPHS PER PERSON ON FIW. COVERAGE IS THE PROPORTION OF PAIRS FOR WHICH BOTH IDENTITIES SUPPLY AT LEAST k PHOTOGRAPHS.

Photographs	ROC-AUC	Gain vs. 1	Coverage
1	0.7339	+0.0000	100%
2	0.7744	+0.0405	100%
3	0.7832	+0.0493	100%
4	0.7923	+0.0584	100%
5	0.7951	+0.0612	99%
6	0.7978	+0.0639	99%
8	0.8003	+0.0664	99%
10	0.8042	+0.0702	97%

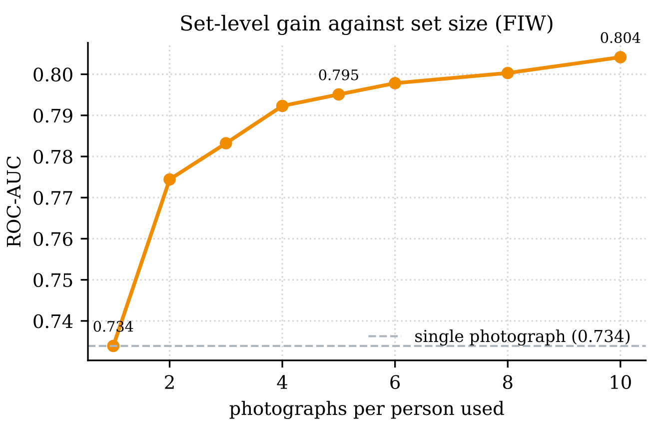

Fig. 8. Set-level ROC-AUC on FIW against the number of photographs aggregated per person. Most of the available gain is realised by the fifth photograph. Computed on frozen embeddings with no training, so it measures the representation rather than model capacity.

TABLE XI

ACCURACY AND COST AGAINST PHOTOGRAPHS PER PERSON. ROC-AUC IS MEASURED ON FIW; TIMINGS ARE PER PAIR ON ONE CPU CORE AT 512 DIMENSIONS. ACCURACY IS FLAT BEYOND FIVE PHOTOGRAPHS WHILE COST CONTINUES TO GROW, WHICH MAKES FIVE A NATURAL OPERATING POINT.

Photos	ROC-AUC	Gain	ms/pair	pairs/s	Slowdown
1	0.7339	+0.0000	0.078	12763	1.0×
2	0.7744	+0.0405	0.099	10078	1.3×
5	0.7951	+0.0612	0.117	8539	1.5×
10	0.8042	+0.0702	0.153	6544	2.0×
20	—	—	0.215	4643	2.7×
40	—	—	0.416	2402	5.3×
80	—	—	1.502	666	19.2×
160	—	—	3.132	319	40.0×
320	—	—	10.718	93	136.8×

D. The operating point: accuracy against cost

Person-level scoring compares two photograph *sets*, so its cost grows with set size while its accuracy saturates. Where those two curves cross is a deployment question that the accuracy curve alone does not answer, and we report it because practitioners must choose how many photographs to collect.

Table XI places the two measurements side by side. Accuracy rises steeply to five photographs, capturing $+0.061$ of the $+0.070$ available at ten, and is essentially flat thereafter. Cost behaves in the opposite way: it is negligible to about twenty photographs and then grows steadily, reaching a $137\times$ slowdown at 320.

The practical consequence is a clear recommendation. **Five photographs per person is close to optimal on both axes**: it captures 87% of the available accuracy gain at $1.5\times$ the single-photograph cost. Collecting more than roughly twenty buys no measurable accuracy while making the system progressively slower.

We note also that the cost curve is a property of the implementation rather than of the method, and that the distinction matters. An earlier version of our code computed the within-set spread by materialising the full Gram matrix and extracting its upper triangle with an index array. That allocation dominated the call at large set sizes: $46\,\mathrm{ms}$ of a $48\,\mathrm{ms}$ call at $n = 160$. Because the required quantity is the mean off-diagonal entry, and for unit-norm rows $\sum_{i\neq j}\langle x_i, x_j\rangle = \|\sum_i x_i\|^2 - n$, it can be computed without forming the matrix at all. Doing so reduced the $n = 160$ cost from $89.8\,\mathrm{ms}$ to $3.1\,\mathrm{ms}$, a $29\times$ improvement, and removed a discontinuity that had made the empirical growth appear super-quadratic. The values are unchanged; the equivalence is asserted by test. We mention this because a cost curve measured on an unoptimised implementation would have overstated the method's true scaling.

E. Score distributions and operating points

Figures 9 and 10 show the underlying distributions rather than only their summaries. The distributions overlap substantially on every corpus — the concrete form of the roughly one-in-four error rate — and the degree of overlap differs enough between corpora that a single global threshold is inappropriate. We therefore calibrate per corpus on a group-disjoint validation slice; a single shared threshold costs between 0.7 and 5.9 accuracy points depending on the corpus.

X. Do the Findings Depend on the Backbone?

All results so far use a single frozen embedding, which bounds what they can claim: a finding could be a property of the task, or an artefact of FaceNet. We therefore repeat the central experiments with ArcFace (`buffalo_l`, w600k_r50) substituted for FaceNet and everything else held fixed — same folds, same negatives, same classifier, same protocol. The two backbones are trained with different objectives on different corpora, so agreement between them is meaningful evidence.

Three observations follow from Table XII.

Identity leakage is backbone-invariant by construction. It is a property of how pairs are partitioned, not of how images are embedded, so the 100% figure of Section IV carries over unchanged. We state this for completeness rather than as a finding.

The person-level gain replicates and strengthens. FaceNet gains $+0.0763$ ROC-AUC on FIW from aggregating photographs; ArcFace gains $+0.0976$. That the effect is larger under the stronger backbone is consistent with its proposed mechanism — aggregation averages away per-photograph nuisance variation, and a better embedding has proportionally more signal left to recover once that variation is suppressed.

The degradation guarantee holds under both. KinFaceW-I, KinFaceW-II and TSKinFace store one photograph per person, and set-level scoring reproduces the single-image result exactly on all three under both backbones. This is a correctness property of the implementation rather than an empirical result, and it is what allows a single system to be deployed across corpora with differing photograph availability.

ArcFace is the stronger backbone in absolute terms on three of four corpora, most markedly on TSKinFace (0.8460 against 0.7326). We do not read this as a contribution: it reflects a better face representation, not a better kinship method, and it is reported so that the backbone-invariance claims above are interpretable.

A. The negative result under a second backbone

Table XIII re-tests the density-fidelity formulation. Across eight backbone–corpus combinations the largest effect in either direction is 0.0025 ROC-AUC, and seven of the eight are negative. The failure is therefore not specific to the embedding it was first measured on, which supports the structural explanation of Section XI: a density matrix estimated from a median of five observations is dominated by its leading eigenvector regardless of which encoder produced those observations.

XI. A Negative Result for Quantum-Inspired Metrics

A. Formulations tested

We evaluated nine formulations, chosen to span the three roles a quantum component can occupy:

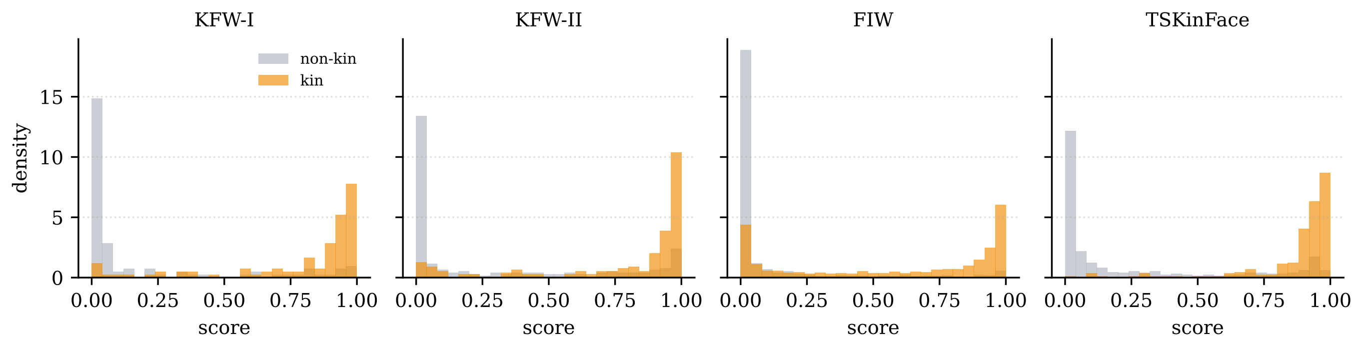

Fig. 9. Kin and non-kin score distributions on the held-out fold of each corpus. The overlap, rather than any summary statistic, is what the reported ROC-AUC measures.

TABLE XII
PERSON-LEVEL AGGREGATION UNDER TWO BACKBONES. THE GAIN REPLICATES AND IS IN FACT LARGER UNDER ARCFACE; THE THREE SINGLE-PHOTOGRAPH CORPORA ARE UNCHANGED TO MACHINE PRECISION UNDER BOTH, AS THE DEGRADATION GUARANTEE REQUIRES.

	FaceNet			ArcFace		
Corpus	Single	Set-level	Δ	Single	Set-level	Δ
KinFaceW-I	0.7588	0.7588	+0.0000	0.7540	0.7540	+0.0000
KinFaceW-II	0.7770	0.7770	+0.0000	0.8190	0.8190	+0.0000
FIW	0.7491	0.8254	+0.0763	0.7764	0.8740	+0.0976
TSKinFace	0.7326	0.7326	+0.0000	0.8460	0.8460	+0.0000

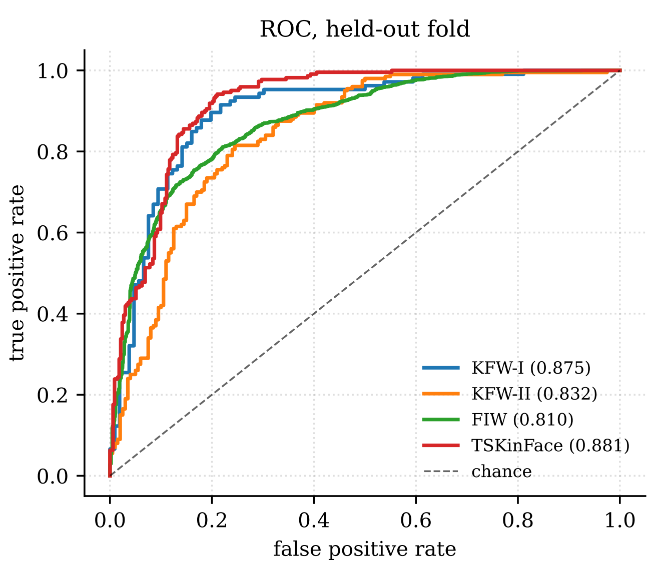

Fig. 10. ROC curves on the held-out fold of each corpus, with grouped k-fold mean ROC-AUC in parentheses.

TABLE XIII
CHANGE IN ROC-AUC FROM ADDING THE DENSITY-FIDELITY ARM $\mathrm{Tr}(\rho_1\rho_2)$ TO THE SET-LEVEL FEATURES. THE FORMULATION FAILS UNDER BOTH BACKBONES ON ALL FOUR CORPORA.

Corpus	FaceNet	ArcFace
KinFaceW-I	-0.0025	-0.0006
KinFaceW-II	-0.0003	$+0.0001$
FIW	-0.0003	-0.0007
TSKinFace	$+0.0001$	-0.0014

TABLE XIV
QUANTUM-INSPIRED FORMULATIONS AGAINST MATCHED CLASSICAL CONTROLS.

Formulation	Control	Outcome
SWAP fidelity (5 seeds)	classical head	$p = 0.945$
SWAP fidelity (20 folds)	classical head	$p = 0.982$
Density fidelity $\mathrm{Tr}(\rho_1\rho_2)$	mean-pooling	-8.7 AUC
Interference phase sweep	classical mixture	$+0.1$ AUC
Entanglement entropy	cosine	0.514 alone
POVM head	capacity-matched	-5.5 AUC
Purity regulariser	decorrelation	-0.9, $p = 0.047$

a similarity measure, a feature extractor, and an inductive bias. Each is compared against a control matched for capacity or intent (Table XIV).

Under grouped k-fold with 20 paired observations, enabling the SWAP-test branch yields 75.66% against 76.34% with it disabled (accuracy $p = 0.158$; ROC-AUC $p = 0.982$). The branch also costs 6–14$\times$ runtime.

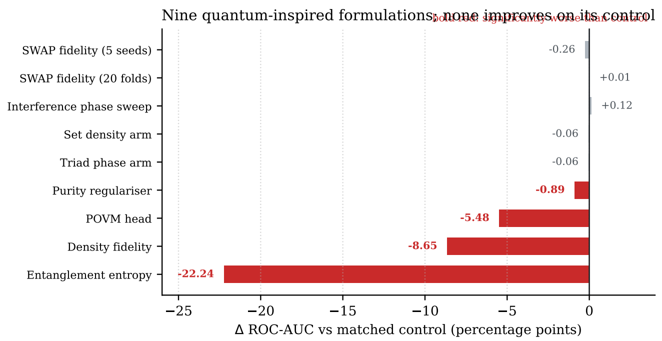

Fig. 11. Nine quantum-inspired formulations against classical controls matched for capacity or intent. None improves on its control; four are significantly worse. Bars are ΔROC-AUC in percentage points.

B. A cautionary observation

The purity regulariser initially measured $+1.3$ ROC-AUC on a single seed, with reduced variance — a result consistent with effective regularisation. Repeated across nine paired runs the sign reversed, and the method proved significantly *worse* than no regulariser (-0.0089, $p = 0.047$). We report this because the single-seed reading was of the kind that is readily published, and it illustrates why capacity-matched controls and repeated trials are necessary when evaluating claims of this type.

C. Two further formulations

The set-level and triadic representations each admit a natural quantum reading, and we tested both as pre-registered controls.

A photo set is a mixed state $\rho = \frac{1}{N}\sum_i |\psi_i\rangle\langle\psi_i|$, and $\mathrm{Tr}(\rho_1\rho_2)$ is the corresponding fidelity. Added to the set features it changes ROC-AUC by -0.0006, -0.0005, $+0.0001$ and -0.0006 on the four benchmarks.

A child explained by two parents is naturally written as a coherent superposition with a relative phase, of which the classical convex mixture is the zero-phase special case. Sweeping the phase changes ROC-AUC by -0.0006 ($p = 0.147$).

This brings the total to nine formulations, none of which improves on its classical control.

All nine share a property that the next subsection identifies as decisive: each compresses the quantum state to a single scalar before the decision is made.

D. Why the approach fails

The failures are structural rather than implementational. Interference-based formulations reduce to their classical counterparts because ℓ_2-normalised FaceNet embeddings carry no phase structure to exploit; the phase sweep in Table XIV is maximised at the classical mixture. Density-matrix formulations require estimating a $d \times d$ operator from the images available for one identity — a median of five — so the resulting ρ is dominated by its leading eigenvector, which is the mean, while accumulating noise from tail directions that are not estimable. This is why $\mathrm{Tr}(\rho_1\rho_2)$ scores *below* simple mean-pooling.

A third factor cuts across both, and bounds what the present negative result licenses. Every formulation above reads its decision out of the quantum state as one number. In a companion study we hold a variational circuit, its folds, its optimiser and its parameter budget fixed and vary only that readout: replacing scalar fidelity with a per-qubit expectation vector improves ROC-AUC by $+0.144$ on FaceNet and $+0.115$ on ArcFace, positive in all eight backbone $\times$ corpus combinations. The information is present in the state and the scalar discards it. We therefore scope the claim of this section to scalar-readout formulations. It does not follow, and we do not claim, that quantum-inspired representations are uninformative for this task; nor does the companion result establish a quantum advantage, since classical controls matched on readout width match or exceed the circuit on the largest corpus.

E. An efficiency contribution

Independently of the above, we contribute an exact reformulation of the simulated SWAP test. The reference implementation materialises a 2^n statevector and permutes all n axes once per qubit, incurring kernel-launch overhead that leaves the accelerator idle. Applying rotations by reshaping to $(B, \text{left}, 2, \text{right})$ and precomputing the CNOT chain and R_z layer as a single diagonal phase vector yields a $380\times$ speedup on GPU ($192 \rightarrow 73{,}094$ pairs/s) with identical output and gradients, verified by test. We note that a closed-form product over qubits is *not* available here: the shared CNOT chain correlates the R_z phases, so the overlap does not factorise. The gain arises from removing overhead, not from approximation.

This is a quantum-versus-quantum speedup. Simulating a circuit is necessarily more costly than the classical operation it parallels, and we measure the quantum branch at 5–14$\times$ slower than the classical path at every qubit count from 2 to 12.

XII. Limitations

Small test sets. KinFaceW-I and KinFaceW-II yield 212 and 400 test pairs per fold, giving confidence intervals of roughly ± 2 points. FIW and TSKinFace are firmer.

Single backbone. All results use frozen FaceNet embeddings. Whether the biases we quantify interact with backbone choice is untested.

Set-level gains are confined to one benchmark. Only FIW stores multiple photographs per identity, so the $+0.0771$ ROC-AUC of Section VIII is measured on a single corpus; the other three are unchanged by construction. Whether the effect holds at other set sizes, and on corpora we could not test, is open.

Negative results are bounded. We tested nine formulations; we do not claim no quantum-inspired method can help on this task, only that these nine, covering the three natural roles, do not. The bound is sharper than a general caution: all nine read the state out as a scalar, and a companion study shows that changing only that readout recovers a large effect. A negative result obtained through a scalar readout is confounded with the readout, and should not be generalised to the underlying representation.

XIII. Conclusion

We have argued that facial kinship verification is conventionally posed and measured in ways that both understate and overstate what the data supports. Pair-level splitting leaks identities into the test set at a rate we measure at 100% on FIW; grouped k-fold with per-fold negative regeneration removes this and reduces seed-to-seed variation from 22.5 to 12.1 points. Against that protocol, treating verification as a person-level rather than photograph-level decision — standard in template-based face recognition, but not previously applied here — is worth $+0.077$ ROC-AUC on the one benchmark that supplies photograph sets, and by construction changes nothing on those that do not. Scoring parent–parent–child triads jointly adds $+0.032$ on TSKinFace.

We also report that nine quantum-inspired formulations, each against a capacity-matched classical control, fail to improve on their controls, with four significantly worse; the causes are structural rather than implementational. All code is released so that the protocol can be adopted and every claim checked.

One limitation bounds these results: set-level gains are measured on a single corpus, because only FIW stores multiple photographs per identity. We note also that the per-relation analysis of Section IX-B found no material difference between relation types once no family was permitted to dominate a fold, so we make no claim that any relation is intrinsically harder.

Reproducibility

Every numeric claim derives from JSON artefacts in `results/honest/`. The protocol is implemented in `src/splits.py`, `src/kfold.py`, and `src/ts_pairs.py`; results are regenerated by `scripts/evaluation/run_kfold.py`. The implementation is covered by 61 tests, including assertions that no group spans a fold boundary and that the fast SWAP-test reformulation matches the reference simulator in both value and gradient.

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